Student number_____ Name_____ Major_____ Grade_____

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Stony Brook Institute at Anhui University

The first semester of 2020—2021 year

Applied Calculus I Final Examination (B)

(Closed book Time 120 minutes)

Question number	I	II	III	IV	V	Total Score
Score						
Marker						

I. Completions (5 items, 3 points each)

Score	

1. $\lim\limits_{n\to\infty} 2^n \sin\dfrac{x}{2^n} =$ _____.

2. If $\lim\limits_{x\to\infty}\left(1-\dfrac{1}{x}\right)^{kx} = e^3$, then $k =$ _____.

3. If function $f(x) = x(x+1)(x+2)\cdots(x+n)(n\geq 2)$,then $f'(0) =$ _____.

4. The monotone decreasing interval of $f(x) = (x-1)\sqrt[3]{x^2}$ is_____.

5. If $y = 1 + xe^y$, then $y' =$ _____.

II. Choices (5 items, 3 points each)

Score	

6. Which of the following functions is odd ( )

A. $y = \dfrac{e^x + e^{-x}}{2}$. B. $y = \dfrac{x^2(x-2)}{x-2}$.

C. $y = \ln(x + \sqrt{x^2+1})$. D. $y = \arcsin x + \arccos x$.

7. If $f(x)=\begin{cases}\frac{2}{3}x^2, & x\le 1\\ x^2, & x>1\end{cases}$, then for the function $f(x)$ at point $x=1$,

( )

A. Its left and right derivatives both exist.
B. Its left derivative exists, right derivative doesn't exist.
C. Its right derivative exists, left derivative doesn't exist.
D. Its left and right derivatives both don't exist.

8. If $x\to 0, e^x-(ax^2+bx+1)=o(x^2)$, then ( )

A. $a=\frac{1}{2}, b=1.$ B. $a=1, b=1.$

C. $a=-\frac{1}{2}, b=1.$ D. $a=-1, b=1.$

9. Let $f(x)=\left|x^{\frac{1}{3}}\right|$, then the point $x=0$ is ( )

A. point of discontinuity. B. inflection point.
C. the maximum point. D. the minimum point.

10. Let $\begin{cases}x=a\cos t\\ y=b\sin t\end{cases}$, then $y''=$ ( )

A. $-\frac{b}{a}\csc^2 t$. B. $\frac{b}{a}\csc^2 t$.

C. $-\frac{b}{a^2\sin^3 t}$. D. $\frac{b}{a^2\sin^3 t}$.

III. Calculations (6 items, 8 points each)

Score	

11. Calculating Derivative $y=\ln\left(e^x+\sqrt{1+e^{2x}}\right)$.

12. Let $y = x^2 \cos x$,Calculating higher-order derivative of $y^{(20)}$.

13. Calculating $\lim\limits_{x \to 0} \dfrac{\tan x - \sin x}{\sin^3 x}$.

14. Suppose that $\lim\limits_{x \to \infty} f'(x) = k$, calculating $\lim\limits_{x \to \infty} \left[f(x+a) - f(x) \right]$.

15. Calculating $\lim_{x\to\infty}\left[\frac{\left((a_1)^{\frac{1}{x}}+(a_2)^{\frac{1}{x}}+\cdots+(a_n)^{\frac{1}{x}}\right)}{n}\right]^{nx}$, $(a_1, a_2, \cdots, a_n > 0)$.

16. Use the third order Taylor formula to approximate the value of $\sqrt[3]{30}$ and estimate the error.

Score	

IV. Proofs（2 items, 7 points each）

17. If the function $f(x)$ is continuous in $[0,1]$, and $f(0)>0, f(1)<1,$ prove that there exists $\xi \in (0,1),$ such that $f(\xi)=\xi^2$.

18. If $x>0$,then try to prove that $\sin x > x-\frac{x^3}{3!}$.

V. Modelling Problem（8 points）

Score	

19. Suppose that the manufacturing cost of a backpack is 40 RMB yuan. If the selling price of a backpack is x RMB yuan, the backpack's demand function is given by $n=\frac{a}{x-40}+b(80-x)$, where a,b are the positive numbers. How much is the optimal selling price when the manufacturers want to get the maximum profit?